Zixiao Wang (王子啸)

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Hong Kong SAR, China

Biography

Zixiao Wang received his Ph.D. in Computer Science and Engineering from The Chinese University of Hong Kong in 2026, under the supervision of Prof. Bei Yu and Prof. Farzan Farnia. He received his M.Sc. in Computer Science and Technology and B.Eng. in Automation (with a second major in Economics) from Tsinghua University.
He will join the Microelectronics Thrust, Function Hub, The Hong Kong University of Science and Technology (Guangzhou) as an Assistant Professor in January 2027.
His research interests primarily lie at the intersection of Electronic Design Automation (EDA) and Machine Learning. His work encompasses areas such as 1) design for manufacturability, 2) simulation within EDA, and 3) the development of hardware-efficient machine learning algorithms.

Education

- **The Chinese University of Hong Kong** — *Aug 2022 – Jun 2026*
  Ph.D. in Computer Science and Engineering — Hong Kong, China
  - Advisor: Prof. Bei Yu & Prof. Farzan Farnia
- **Tsinghua University** — *Sep 2019 – Jun 2022*
  M.Sc. in Computer Science and Technology — Beijing, China
- **Tsinghua University** — *Sep 2015 – Jun 2019*
  B.Eng. in Automation — Beijing, China
- **Tsinghua University** — *Sep 2016 – Jun 2019*
  B.A. in Economics (Second Major) — Beijing, China

Selected Awards

Full Postgraduate Studentship	CUHK	2022–2026
Outstanding Graduate of Computer Science and Technology	Tsinghua University	2022
Qi Hang Scholarship	Tsinghua University	2019
Academician Ni Weidou Scholarship Endowment Fund	Tsinghua University	2018
Zheng Geru Scholarship	Tsinghua University	2017
1st runner-up in MLCAD Contest	MLCAD	2026
2nd Place Award in Technical Challenge (Team Vulcan)	RoboCup	2018
3rd Place Award in 1v1 Contest (Team Vulcan)	RoboCup	2018

Experience

- **Primarius Technologies** — *Jan 2026 – Jun 2026*
  Research Intern — Shanghai, China
- **Huawei Noah's Ark Lab** — *Aug 2024 – May 2025*
  Research Intern — Hong Kong, China
- **SenseTime Research** — *Apr 2023 – Dec 2023*
  Research Intern — Hong Kong, China
- **Tencent AI Lab** — *Oct 2020 – Aug 2022*
  Research Intern — Shenzhen, China
- **UBTECH Robotics** — *Sep 2017 – Aug 2018*
  Research Intern — Beijing, China

PUBLICATIONS

Conference Papers

[C17] **Zixiao Wang**, Leilei Jin, Zhen Zhuang, Rongmei Chen, Bei Yu, "CSCO: A Backside-PDN-Aware Clock–Signal Co-Optimization Framework for Improved PPA", IEEE/ACM International Conference on Computer-Aided Design (**ICCAD**), San Jose, Nov. 08–12, 2026.

[C16] Chaofan Wang, Ziyang Yu, Su Zheng, **Zixiao Wang**, Yifan Shi, Bei Yu, "Moreau-ILT: Efficient Inverse Lithography Exploiting Weak Convexity", IEEE/ACM International Conference on Computer-Aided Design (**ICCAD**), San Jose, Nov. 08–12, 2026.

[C15] Leilei Jin, Haoyang Xu, Siyuan Liang, Zhen Zhuang, Zhou Hu, **Zixiao Wang**, Bei Yu, Rongmei Chen, Tsung-Yi Ho, "BSPDN-Elite: A Comprehensive Framework for Optimizing Timing, Power and Routing Resources in BSPDN Designs", ACM/IEEE Design Automation Conference (**DAC**), Long Beach, Jul. 26–29, 2026.

[C14] **Zixiao Wang**, Jieya Zhou, Xinyun Zhang, Shoubo Hu, Farzan Farnia, Bei Yu, "DiffResist: Physics-Constrained Diffusion for Photoresist Modeling", IEEE/ACM Proceedings Design, Automation and Test in Europe (**DATE**), Verona, Italy, Apr. 20–22, 2026. [paper] [poster] [slides]

[C13] **Zixiao Wang**, Tianshu Hou, Chenghan Wang, Zhen Zhuang, Tsung-Yi Ho, Farzan Farnia, Bei Yu, "FastRW: An Efficient Random Walk Method for Steady-State Thermal Analysis", IEEE/ACM Proceedings Design, Automation and Test in Europe (**DATE**), Verona, Italy, Apr. 20–22, 2026. [paper] [poster] [slides] [code]

[C12] Wenbo Xu, Silin Chen, Yibo Huang, Xinyun Zhang, **Zixiao Wang**, Bei Yu, Ningmu Zou, "Node2Node: Node Adaptation with Transformer for Cross-Node Hotspot Detection", IEEE/ACM Proceedings Design, Automation and Test in Europe (**DATE**), Verona, Italy, Apr. 20–22, 2026. [paper]

[C11] Rongliang Fu, Libo Shen, Ziyi Wang, Zhengxing Lei, **Zixiao Wang**, Junying Huang, Bei Yu, Tsung-Yi Ho, "DCLOG: Don't Cares-based Logic Optimization using Pre-training Graph Neural Networks", IEEE/ACM Asian and South Pacific Design Automation Conference (**ASPDAC**), Hong Kong, Jan. 19–22, 2026. [paper]

[C10] Xinyun Zhang*, **Zixiao Wang***, Yurui Kuang, Bei Yu, "Video-based Visible-Event Cross-modal Person Re-identification for Edge AI Surveillance Systems", IEEE/ACM Asian and South Pacific Design Automation Conference (**ASPDAC**), Hong Kong, Jan. 19–22, 2026. [paper] [slides]

[C9] **Zixiao Wang**, Farzan Farnia, Zhenghao Lin, Yunheng Shen, Bei Yu, "On the Distributed Evaluation of Generative Models", IEEE/CVF International Conference on Computer Vision Workshops (**ICCVW**), Honolulu, Oct. 19–20, 2025, pp. 7703–7712. [paper] [arXiv]

[C8] Ziyang Yu, Peng Xu, **Zixiao Wang**, Binwu Zhu, Qipan Wang, Yibo Lin, Runsheng Wang, Bei Yu, Martin Wong, "SDM-PEB: Spatial-Depthwise Mamba for Enhanced Post-Exposure Bake Simulation", ACM/IEEE Design Automation Conference (**DAC**), San Francisco, Jun. 22–25, 2025. [paper] [poster] [slides]

[C7] **Zixiao Wang**, Junwu Weng, Mengyuan Liu, Bei Yu, "FlexPose: Pose Distribution Adaptation with Limited Guidance", AAAI Conference on Artificial Intelligence (**AAAI**), Philadelphia, Feb. 25–Mar. 04, 2025. [paper] [poster]

[C6] **Zixiao Wang**, Jieya Zhou, Su Zheng, Shuo Yin, Kaichao Liang, Shoubo Hu, Xiao Chen, Bei Yu, "TorchResist: Open-source Differentiable Resist Simulator", SPIE Advanced Lithography + Patterning, San Jose, Feb. 23–27, 2025. [paper] [poster] [code]

[C5] **Zixiao Wang***, Yunheng Shen*, Xufeng Yao, Wenqian Zhao, Yang Bai, Farzan Farnia, Bei Yu, "ChatPattern: Layout Pattern Customization via Natural Language", ACM/IEEE Design Automation Conference (**DAC**), San Francisco, Jun. 23–27, 2024. [paper] [poster] [slides]

[C4] Yang Bai, Wenqian Zhao, Shuo Yin, **Zixiao Wang**, Bei Yu, "ATFormer: A Learned Performance Model with Transfer Learning Across Devices for Large-Scale Transformers", Empirical Methods in Natural Language Processing (**EMNLP**), Singapore, Dec. 06–10, 2023. [paper] [poster] [slides]

[C3] **Zixiao Wang***, Yunheng Shen*, Wenqian Zhao, Yang Bai, Guojin Chen, Farzan Farnia, Bei Yu, "DiffPattern: Layout Pattern Generation via Discrete Diffusion", ACM/IEEE Design Automation Conference (**DAC**), San Francisco, Jul. 09–13, 2023. [paper] [poster] [slides]

[C2] **Zixiao Wang**, Junwu Weng, Chun Yuan, Jue Wang, "Truncate-Split-Contrast: A Framework for Learning from Mislabeled Videos", AAAI Conference on Artificial Intelligence (**AAAI, Oral**), Washington, Feb. 7–14, 2023. [paper] [poster] [slides] [code]

[C1] Haijin Ding, **Zixiao Wang**, Rebing Wu, "Enhancing the security of multi-agent networked control systems using QKD based homomorphic encryption", IEEE Conference on Decision and Control (**CDC**), Miami Beach, Dec. 17–19, 2018. [paper]

Journal Papers

[J5] **Zixiao Wang**, Wenqian Zhao, Yunheng Shen, Yang Bai, Guojin Chen, Farzan Farnia, Bei Yu, "DiffPattern-Flex: Efficient Layout Pattern Generation via Discrete Diffusion", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), vol. 44, no. 12, pp. 4679–4690, 2025. [paper]

- **[J4]** Wenqian Zhao, Lancheng Zou, **Zixiao Wang**, Xufeng Yao, Bei Yu, "HAPE: Hardware-Aware LLM Pruning For Efficient On-Device Inference Optimization", ACM Transactions on Design Automation of Electronic Systems (**TODAES**), vol. 30, no. 04, pp. 61:1–61:18, 2025. [paper] (**Editor's Pick**)
- **[J3]** Wenqian Zhao, Shuo Yin, Chen Bai, **Zixiao Wang**, Bei Yu, "BAQE: Backend-Adaptive DNN Deployment via Synchronous Bayesian Quantization and Hardware Configuration Exploration", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), vol. 44, no. 04, pp. 1394–1405, 2025. [paper]
- **[J2]** Guojin Chen, **Zixiao Wang**, Bei Yu, David Z. Pan, Martin D.F. Wong, "Ultrafast Source Mask Optimization via Conditional Discrete Diffusion", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), vol. 43, no. 07, pp. 2140–2150, 2024. [paper]
- **[J1]** Yang Bai, Xufeng Yao, Qi Sun, Wenqian Zhao, Shixin Chen, **Zixiao Wang**, Bei Yu, "GTCO: Graph and Tensor Co-Design for Transformer-based Image Recognition on Tensor Cores", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), vol. 43, no. 02, pp. 586–599, 2024. [paper]

Software

- **FastRW** *2026*
  GPU random-walk thermal solver and experiment harness for the FastRW / FasterRW algorithms (Apple Metal + C++17).
- **TorchResist** *2025*
  An open-source differentiable photoresist simulator that is implemented with Pytorch.
- **MoreauPruner** *2024*
  Robust pruning framework for large language models against weight perturbations. [paper]

Professional Service

- **Technical Program Committee Member**
  IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), 2027
- **Journal Reviewer**
  Transactions on Machine Learning Research (TMLR); IEEE Transactions on Consumer Electronics (TCE)
- **Conference Reviewer**
  CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, and AAAI

Teaching Experience

- **CSCI3320** *2023-R2*
  Fundamentals of Machine Learning, CUHK
- **ENGG2760A** *2023-R1*
  Probability for Engineers, CUHK
- **CSCI5030** *2022-R2*
  Machine Learning Theory, CUHK
- **ENGG2020** *2022-R1*
  Digital Logic and Systems, CUHK